

1.3 Homework Solutions: Measurement quiz review

Notation: $=$ joins values that are exactly equal; $\approx$ introduces a rounded value. Students should write it the same way — once you round the calculator display, the equals sign is no longer true. Each conversion runs as one chain of unit factors and is evaluated once, so there is nothing to round in the middle.

1. Significant digits.

(a) **3**

(c) **3**

(e) **4**

(b) **2**

(d) **4**

(f) **4**

(b) The leading zeros in 0.0078 only locate the decimal point; they are not measured.
(d) The trailing zero in 250.0 is after the decimal point, so it was measured — that is the whole reason for writing it. (e) 0.01070 has leading zeros that do not count and a trailing zero that does. (f) Scientific notation settles the question for good: everything in the coefficient counts, and nothing else does.

2. Round to three significant figures.

(a) $\sqrt{7} = 2.6457513 \dots \approx \mathbf{2.65}$

(b) 384,400 km $\approx \mathbf{384,000 \text{ km}}$, better written $\mathbf{3.84 \times 10^5 \text{ km}}$

The third figure is the 4 in the thousands place; everything after it goes to zero. Writing 384,000 is ambiguous — it looks like it could have six figures — which is exactly why scientific notation is the safer form.

3. Scientific notation.

(a) 299,792,458 m/s $= 2.99792458 \times 10^8 \text{ m/s} \approx \mathbf{3.00 \times 10^8 \text{ m/s}}$

Write 3.00, not 3. The two trailing zeros are the significant figures you were asked for.

(b) 0.000000532 m $= \mathbf{5.32 \times 10^{-7} \text{ m}}$ (already exactly three figures — nothing to round)

4. Ordinary decimal numbers.

(a) $6.37 \times 10^6 \text{ m} = \mathbf{6,370,000 \text{ m}}$

(b) $6.7 \times 10^{-2} \text{ s} = \mathbf{0.067 \text{ s}}$

5. Reading the ruler. The bar runs from 0 to a little past the 7.4 cm mark.

(a) **7.45 cm** — the 7.4 is certain (the end lies between the 4th and 5th millimeter marks past 7), and the final 5 is the estimated digit, judged by eye as about halfway between them.

Accept 7.44 to 7.46. Do not accept 7.4 (no estimated digit — it throws away information the ruler gives you) or 7.450 (an estimate of an estimate).

(b) **Three significant figures.**

$$(c) \quad 7.45 \text{ cm} \times \frac{10 \text{ mm}}{1 \text{ cm}} = \mathbf{74.5 \text{ mm}} \qquad 7.45 \text{ cm} \times \frac{1 \text{ m}}{100 \text{ cm}} = \mathbf{0.0745 \text{ m}}$$

All three are the same measurement and all three have three significant figures. Changing units never changes the precision — the leading zeros in 0.0745 are placeholders, not measured digits.

6. Convert.

$$(a) \quad 1776 \text{ ft} \times \frac{1 \text{ m}}{3.28 \text{ ft}} = \frac{1776}{3.28} \text{ m} = 541.46 \dots \approx \mathbf{541 \text{ m}}$$

The published height is 541.3 m, so this checks. Note that 1776 ft is not a measurement at all — the height was chosen to match the year, so the number is exact by design and every digit means something. The 3.28 is the rounded factor here, and it is what limits the answer to three figures.

$$(b) \quad 40 \text{ nickels} \times \frac{5.000 \text{ g}}{1 \text{ nickel}} \times \frac{1 \text{ kg}}{1000 \text{ g}} = \mathbf{0.2000 \text{ kg}}$$

Keep all four figures: the mint specification is exact to four decimal places of a gram and the count of 40 is exact, so nothing here forces rounding. This is why a nickel makes a free calibration check for a classroom balance.

7. Marathon record: 42.195 km in 2 h 00 min 35 s.

$$\text{First the time, in seconds: } 2 \text{ h} \times \frac{3600 \text{ s}}{1 \text{ h}} + 35 \text{ s} = 7200 \text{ s} + 35 \text{ s} = 7235 \text{ s}$$

$$v = \frac{d}{t} = \frac{42,195 \text{ m}}{7235 \text{ s}} = 5.8320 \dots \approx \mathbf{5.83 \text{ m/s}}$$

$$v = \frac{42.195 \text{ km}}{7235 \text{ s}} \times \frac{3600 \text{ s}}{1 \text{ h}} = 20.995 \dots \approx \mathbf{21.0 \text{ km/h}}$$

Both start from the same measured distance and time. Converting the 5.83 m/s instead is fine too ($\times 3.6$), as long as you carry the unrounded 5.8320 into it. Worth saying out loud: 21 km/h is faster than most students can sprint, and the record holder keeps it up for two hours.

8. Frecciarossa at 300 km/h.

$$(a) \quad 300 \frac{\text{km}}{\text{h}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = \frac{300,000}{3600} \frac{\text{m}}{\text{s}} = 83.33 \dots \approx \mathbf{83.3 \text{ m/s}}$$

$$(b) \quad 300 \frac{\text{km}}{\text{h}} \times \frac{1 \text{ mi}}{1.61 \text{ km}} = \frac{300}{1.61} \frac{\text{mi}}{\text{h}} = 186.33 \dots \approx \mathbf{186 \text{ mi/h}}$$

Rome to Milan is about 570 km and the fast service takes roughly 3 hours, so the average speed is near 190 km/h — well under the 300 the train is built for. Stops, station approaches, and older track at each end account for the difference. Top speed and average speed are different quantities.

9. Badminton and baseball.

(a) Badminton: $493 \frac{\text{km}}{\text{h}} \times \frac{1000 \text{ m}}{1 \text{ km}} \times \frac{1 \text{ h}}{3600 \text{ s}} = 136.94 \dots \approx \mathbf{137 \text{ m/s}}$

Fastball: $95 \frac{\text{mi}}{\text{h}} \times \frac{5280 \text{ ft}}{1 \text{ mi}} \times \frac{1 \text{ m}}{3.28 \text{ ft}} \times \frac{1 \text{ h}}{3600 \text{ s}} = \frac{95 \times 5280}{3.28 \times 3600} \frac{\text{m}}{\text{s}} = 42.48 \dots \approx \mathbf{42.5 \text{ m/s}}$

- (b) The neat way needs no metric units at all — the feet and the miles cancel each other:

$$t = \frac{60.5 \text{ ft}}{95 \text{ mi/h}} \times \frac{1 \text{ mi}}{5280 \text{ ft}} \times \frac{3600 \text{ s}}{1 \text{ h}} = \frac{60.5 \times 3600}{95 \times 5280} \text{ s} = 0.4342 \dots \approx \mathbf{0.43 \text{ s}}$$

Equivalently, $t = \frac{18.44 \text{ m}}{42.48 \text{ m/s}} \approx 0.43 \text{ s}$ — same answer, more work.

The hitter has about 0.43 s, and roughly 0.25 s of that is gone before the body can respond at all. Less than two tenths of a second remains to judge the pitch and start the swing — and the decision has to be made while the ball is still only part way to the plate. This is why hitters talk about reading the pitcher's arm rather than watching the ball.

10. Proton to electron.

$$\frac{1.673 \times 10^{-27} \text{ kg}}{9.109 \times 10^{-31} \text{ kg}} = \frac{1.673}{9.109} \times 10^{-27-(-31)} = 0.18366 \dots \times 10^4 = 1836.6 \dots \approx \mathbf{1837}$$

Four significant figures, because both masses are given to four and a quotient carries no more than the least precise value that went into it. The answer is a pure number — the kilograms cancel — so it has no units.

Watch the exponents: subtracting a negative exponent adds. Students who write 10^{-58} have multiplied instead of divided; a quick sanity check is that the proton is obviously the heavier particle, so the answer must come out well above 1.

11. The Atlantic, widening 2.5 cm/yr.

(a) $2.5 \frac{\text{cm}}{\text{yr}} \times \frac{1 \text{ m}}{100 \text{ cm}} \times \frac{1 \text{ yr}}{3.156 \times 10^7 \text{ s}} = \frac{2.5}{100 \times 3.156 \times 10^7} \frac{\text{m}}{\text{s}} = 7.921 \dots \times 10^{-10} \approx \mathbf{7.9 \times 10^{-10} \text{ m/s}}$

Two significant figures — the 2.5 has two. This is a number that only makes sense in scientific notation: written out it is 0.00000000079 m/s, and nobody can read that.

(b) $1.0 \text{ m} \times \frac{100 \text{ cm}}{1 \text{ m}} \times \frac{1 \text{ yr}}{2.5 \text{ cm}} = \mathbf{40 \text{ yr}}$

So the Atlantic has widened by about a meter since these students' parents were born — a rate that is imperceptible per second and obvious per lifetime.

12. Three measurements of the hallway.

(a) $\frac{24.1 + 24.0 + 24.2}{3} \text{ m} = \frac{72.3}{3} \text{ m} = \mathbf{24.1 \text{ m}}$

- (b) The three readings disagree in the tenths place, so the tenths place is already the

uncertain digit — the hallway is 24.1 ± 0.1 m. Writing 24.1000 m claims the length is known to a tenth of a millimeter, which is a thousand times better than the measurements actually agree with each other.

The extra zeros came from dividing by 3 on a calculator, not from the tape measure. Averaging repeated measurements does make the result more reliable, but it cannot invent precision the instrument never had. The correct report is 24.1 m — three significant figures, the same as each reading.

13. Challenge: gasoline.

- (a) Two conversions in one chain — currency and volume:

$$1.82 \frac{\text{€}}{\text{L}} \times \frac{\$1.09}{\text{€1}} \times \frac{3.785 \text{ L}}{1 \text{ gal}} = 1.82 \times 1.09 \times 3.785 \frac{\$}{\text{gal}} = 7.5086 \dots \approx \text{\$7.51 per gallon}$$

Set the factors up so the euros cancel and the liters cancel, leaving dollars over gallons. If the answer comes out near \$0.44, one of the two factors was inverted — and the units in the cancelling will show which.

- (b) $\$7.51 - \$3.45 = \text{\$4.06 more per gallon}$, roughly **2.2 times** the New York price.

Most of that gap is tax, not the cost of the fuel.

(c) $45 \text{ L} \times \frac{\text{€1.82}}{1 \text{ L}} \times \frac{\$1.09}{\text{€1}} = \text{\$89.27}$

Or €81.90 before converting. Money is the one place where rounding to the cent is expected regardless of significant figures — a price is an exact amount, not a measurement.